**14.** The characterization is just like that in Exercise 13 with the requirement that both $G$ and $G'$ be abelian groups.

## 40. Group Presentations

**1.** Three presentations of $\mathbb{Z}_4$ are $(a : a^4 = 1)$,
$(a, b : a^4 = 1, b = a^2)$, and $(a, b, c : a = 1, b^4 = 1, c = a)$.

**2.** Thinking of $a = \rho_1, b = \mu_1$, and $c = \rho_2$, we obtain the presentation $(a, b, c : a^3 = 1, b^2 = 1, c = a^2, ba = cb)$. Starting with this presentation,the relations can be used to express every word in one of the forms $1, a, b, a^2, ab$, or $a^2b$, so a group with this presentation has at most 6 elements. Because the relations are satisfied by $S_3$, we know it must be a presentation of a group isomorphic to $S_3$. (Many other answers are possible.)

**3.** (See the answer in the text.)

**4.** Let $G$ be a nonabelian group of order 14. By Sylow theory, there exists a normal subgroup $H$ of order 7. Let $b$ be an element of $G$ that is not in $H$. Because $G/H \simeq \mathbb{Z}_2$, we see that $b^2 \in H$. If $b^2 \neq 1$, then $b$ has order 14 and $G$ is cyclic and abelian. Thus $b^2 = 1$. Let $a$ be a generator for the cyclic group $H$. Now $bHb^{-1} = H$ so $bab^{-1} \in H$, so $ba = a^r b$ for some value of $r$ where $1 \leq r \leq 6$. If $r = 1$, then $ba = ab$ and $G$ is abelian. By Exercise 13, Part(**b**), the presentation

$$(a, b : a^7 = 1, b^2 = 1, ba = a^r b)$$

gives a group of order $2 \cdot 7 = 14$ if and only if $r^2 \equiv 1 \pmod 7$. Of the possible values $r = 2$, 3, 4, 5, 6, only $r = 6$ satisfies this condition. Thus every nonabelian group of order 14 is isomorphic to the group with presentation $(a, b : a^7 = 1, b^2 = 1, ba = a^6 b)$. We know then that this group must be isomorphic to the dihedral group $D_7$. Of course, $\mathbb{Z}_{14}$ is the only abelian group of order 14.

**5.** Let $G$ be nonabelian of order 21. By Sylow theory, there exists a normal subgroup $H$ of order 7. Let $b$ be an element of $G$ that is not in $H$. Because $G/H \simeq \mathbb{Z}_3$, we see that $b^3 \in H$. If $b^3 \neq 1$, then $b$ has order 21 and $G$ is cyclic and abelian. Thus $b^3 = 1$. Let $a$ be a generator for the cyclic group $H$. Now $bHb^{-1} = H$ so $bab^{-1} \in H$, so $ba = a^r b$ for some value of $r$ where $1 \leq r \leq 6$. If $r = 1$, then $ba = ab$ and $G$ is abelian. By Exercise 13, Part(**b**), the presentation

$$(a, b : a^7 = 1, b^3 = 1, ba = a^r b)$$

gives a group of order $3 \cdot 7 = 21$ if and only if $r^3 \equiv 1 \pmod 7$. Of the possible values $r = 2$, 3, 4, 5, 6, both $r = 2$ and $r = 4$ satisfy this condition. To see that the presentations with $r = 2$ and $r = 4$ yield isomorphic groups, consider the group having this presentation with $r = 2$, and let us form a new presentation of it, taking the same $a$ but replacing $b$ by $c = b^2$. We then have $a^7 = 1$ and $c^3 = 1$, but now $ca = b^2a = a^4b^2 = a^4c$. Thus in terms of the elements $a$ and $c$, this group has presentation $(a, c : a^7 = 1, c^3 = 1, ca = a^4c)$. This shows the two values $r = 2$ and $r = 4$ lead to isomorphic presentations. Thus every group of order 21 is isomorphic to either $\mathbb{Z}_{21}$ or to the group with presentation

$$(a, b : a^7 = 1, b^3 = 1, ba = a^2 b).$$

**6.** The definition is incorrect. A consequence may be any element of the normalizer of the group generated by the relators in the free group on the generators. Also, one should say, "the relators in a presentation of a group".

A **consequence** of the set of relators in a group presentation is any element of the least normal subgroup, containing the relators, of the free group on the generators of the presentation.

**7.** The definition is incorrect. See Example 40.3 and the definition (in bold type within the text) that follows.

Two group presentations are **isomorphic** if and only if the groups $G$ and $G'$ presented by them are isomorphic.

**8.** T T F F F T T F T F (Concerning the answer to Part(**a**), for any group $G$, the presentation $F[G]$ with relators the elements of the kernel of the homomorphism $\phi : F[G] \to G$ where $\phi(g) = g$ for $g \in G$, as described in Theorem 39.12, is a presentation of $G$. Concerning the answer to Part(**j**), the presentation $(a, b, c : c = b)$ is a free group on two generators.)

**9.** Let $G$ be nonabelian of order 15. By Sylow theory, there exists a normal subgroup $H$ of order 5. Let $b$ be an element of $G$ that is not in $H$. Because $G/H \simeq \mathbb{Z}_3$, we see that $b^3 \in H$. If $b^3 \neq 1$, then $b$ has order 15 and $G$ is cyclic and abelian. Thus $b^3 = 1$. Let $a$ be a generator for the cyclic group $H$. Now $bHb^{-1} = H$ so $bab^{-1} \in H$, so $ba = a^r b$ for some value of $r$ where $1 \leq r \leq 4$. If $r = 1$, then $ba = ab$ and $G$ is abelian. By Exercise 13, Part(**b**), the presentation

$$(a, b : a^5 = 1, b^3 = 1, ba = a^r b)$$

gives a group of order $3 \cdot 5 = 15$ if and only if $r^3 \equiv 1 \pmod 5$. But none of $2^3, 3^3$, or $4^3$ is congruent to 1 modulo 5, so there are no nonabelian groups of order 15.

**10.** Exercise 13, Part(**b**), shows that the given presentation is a group of order $2 \cdot 3 = 6$ if and only if $2^2 \equiv 1 \pmod 3$, which is the case. Thus we do have a group of order 6. Because the elements $1, a, b, a^2, ab, a^2b$ are all distinct and because $ba = a^2b$, the group is not abelian, for if $ba = ab$, then $a^2b = ab$ from which we deduce that $a = 1$.

**11.** Let $G$ be nonabelian of order 6. By Sylow theory, there exists a normal subgroup $H$ of order 3. Let $b$ be an element of $G$ that is not in $H$. Because $G/H \simeq \mathbb{Z}_2$, we see that $b^2 \in H$. If $b^2 \neq 1$, then $b$ has order 6 and $G$ is cyclic and abelian. Thus $b^2 = 1$. Let $a$ be a generator for the cyclic group $H$. Now $bHb^{-1} = H$ so $bab^{-1} \in H$, so $ba = a^r b$ for some value of $r$ where $1 \leq r \leq 2$. If $r = 1$, then $ba = ab$ and $G$ is abelian. Thus $ba = a^2$. The preceding exercise shows that the presentation $(a, b : a^3 = 1, b^2 = 1, ba = a^2 b)$ gives a nonabelian group of order 6, and this exercise shows that a nonabelian group of order 6 is isomorphic to one with this presentation. Thus every nonabelian group of order 6 is isomorphic to $S_3$.

**12.** Every element of $A_4$ can be written as a product of disjoint cycles involving some of the numbers 1, 2, 3, 4 and each element is also an even permutation. Because no product of such disjoint cycles can give an element of order 6, we see that $A_4$ has no elements of order 6, and hence no subgroup isomorphic to $\mathbb{Z}_6$, the only possibility for an abelian subgroup of order 6. Therefore, any subgroup of order 6 of $A_4$ must be nonabelian, and hence isomorphic to $S_3$ by the preceding exercise. Now $S_3$ has two elements of order 3 and three elements of order 2. The only *even* permutations in $S_4$ of order 2 are products of two disjoint transpositions, and the only such permutations are (1,2)(3,4) and (1,3)(2,4) and (1,4)(2,3). Thus a subgroup of $A_4$ isomorphic to $S_3$ must contain all three of these elements. It must also contain an element of order 3: we might as well assume that it is the 3-cycle (1, 2, 3). Then it must contain $(1, 2, 3)^2 = (1, 3, 2)$, and the identity would be the sixth element. But this set is not closed under multiplication, for (1, 2)(3, 4)(1, 2, 3) = (2, 4, 3). Thus $A_4$ has no nonabelian subgroup of order 6 either.

**13. a.** We know that when computing integer sums modulo $n$, we may either reduce modulo $n$ after each addition, or add in $\mathbb{Z}$ and reduce modulo $n$ at the end. The same is true for products, as we now show. Suppose $c = nq_1 + r_1$ and $d = nq_2 + r_2$, both in accord with the division algorithm. Then

$$cd = n(nq_1q_2) + n(q_1r_2 + r_1q_2) + r_1r_2,$$

showing that the remainder of $cd$ modulo $n$ is the remainder of $r_1r_2$ modulo $n$. That is, it does not matter whether we first reduce modulo $n$ and then multiply and reduce, or whether we multiply in $\mathbb{Z}$ and then reduce.

Turning to our problem and delaying reduction modulo $m$ and $n$ of sums and products *in exponents* to the end, we have

$$a^sb^t[(a^ub^v)(a^wb^z)] = a^sb^t[a^{u+wr^v}b^{v+z}] = a^{s+(u+wr^v)r^t}b^{t+v+z} \tag{1}$$

and

$$[(a^sb^t)(a^ub^v)]a^wb^z = [a^{s+ur^t}b^{t+v}]a^wb^z. \tag{2}$$

Before we can continue this last computation, we must reduce the exponent $t+v$ modulo $n$, for in the next step $t + v$ will appear as an exponent of an exponent, rather than as a sum or product of first exponents where we are allowed to delay our reduction modulo $n$ to the end. Let $t + v = nq_1 + r_1$ by the division algorithm. Note that because both $t$ and $v$ lie in the range from 0 to $n$ - 1, either $q_1 = 1$ or $q_1 = 0$. Continuing, we see the expression (2) is equal to

$$a^{s+ur^t+wr^{r_1}}b^{t+v+z}. \tag{3}$$

Comparing (1) and (3), we see the associative law holds if and only if

$$s + (u + wr^v)r^t \equiv s + ur^t + wr^{r_1} \pmod{m}$$

and

$$t + v + z \equiv t + v + z \pmod{m}.$$

Of course this second condition is true, and the first one reduces to

$$wr^{v+t} \equiv wr^{r_1} \pmod{m}.$$

Now this relation must hold for all $w$ where $0 \leq w < m$ and for all $v$ and $t$ from 0 to $n$ - 1. Taking $w$ = 1 and $v + t = n$ so that $r_1 = 0$, we see that we must have $r^n \equiv 1 \pmod{m}$. On the other hand, if this is true, then

$$wr^{v+t} = wr^{nq_1+r_1} = w(r^n)^{q_1}r^{r_1} \equiv wr^{r_1} \pmod{m}.$$

This completes the proof.

**b.** Part(**a**) proved the associative law, and $a^0b^0$ is the identity for multiplication. Given $a^ub^v$, we can find $a^sb^t$ such that $(a^sb^t)(a^ub^v) = a^0b^0$ by determining $t$ and $s$ is succession so that

$$t \equiv -v \pmod{n} \quad \text{and} \quad s \equiv -u(r^t) \pmod{m}.$$

The "left group axioms" hold, so we have a group of order $mn$.